

Remote Work and Optimal Congestion Pricing: Evidence from the COVID-19 Demand Shock

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Abstract

This paper quantifies how optimal congestion taxes change when freeway congestion falls in five large California counties: Los Angeles, San Francisco, San Diego, Santa Clara, and Orange. I compare pre-pandemic conditions (2018 to 2019) to post-pandemic conditions (2021 to 2022), using the pandemic era collapse in commuting as a plausibly exogenous shift in travel demand. Combining hourly Caltrans PeMS measures of Vehicle Miles Traveled, the Travel Time Index, and speeds with county year remote work shares and wages from the American Community Survey, I apply a standard externalities framework to estimate marginal external congestion costs, implied Pigouvian taxes per mile, and deadweight loss. Traffic and congestion decline sharply, especially during peak hours, which mechanically pushes optimal taxes and welfare losses downward. Yet median wages rise over the same period, increasing the value of travel time. Once this shift is incorporated, optimal taxes and deadweight loss can increase even as congestion falls. A decomposition separates the change in the optimal tax into a congestion component and a value of time component, showing that in several cases the latter dominates. Finally, the analysis treats the value of time as tied to the median wage, but the results highlight a natural extension: congestion pricing may change the composition of drivers, making the value of time of the marginal driver policy dependent, a margin typically treated as fixed in the optimal congestion taxation literature.

Keywords: Freeway congestion, congestion pricing, optimal congestion taxes, vehicle miles traveled, travel time index, deadweight loss, COVID-19, remote work, welfare analysis

JEL Classification: R40, R41, D62, H43

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1 Introduction

Traffic congestion poses significant economic challenges in urban areas by reducing productivity, increasing travel times, and diminishing overall well-being. This paper studies how the welfare costs of congestion and the implied optimal congestion taxes change when freeway congestion falls in five California counties: Los Angeles, San Francisco, San Diego, Santa Clara, and Orange. I focus on all freeways within these counties.¹ The analysis uses detailed hourly data from the Caltrans Performance Measurement System (PeMS) for the years 2018 to 2022, excluding the peak pandemic year of 2020.

The COVID-19 pandemic generated a large and sudden shock to travel demand through the rise of remote work, changes in commuting behavior, and broader shifts in daily activity patterns. In this paper, the pandemic provides a source of plausibly exogenous variation in traffic volumes and congestion. I compare pre-pandemic traffic conditions (2018–2019) with post-pandemic conditions (2021–2022) to quantify how marginal external congestion costs and optimal congestion taxes respond to this demand shock.

The empirical analysis combines hourly freeway-level data on Vehicle Miles Traveled (VMT) for passenger vehicles and trucks, the Travel Time Index (TTI), and average speeds from PeMS with county-level wages and remote work shares from the American Community Survey (ACS). Building on the framework in [Yang, Purevjav, and Li \(2020\)](#) and [Akbar and Duranton \(2017\)](#), I use these variables within a standard externalities model to compute marginal external congestion costs (MECC), marginal social costs (MSC), implied optimal Pigouvian congestion taxes, and the associated deadweight loss (DWL) before and after the pandemic shock.

The first set of results documents a clear decline in traffic volumes and congestion in the post-pandemic period, especially during peak hours. Holding the value of time fixed, this reduction in congestion mechanically lowers marginal external costs, implied optimal con-

¹ A complete list of included freeways is provided in Table A.1 in the appendix.

gestion taxes, and deadweight loss. However, wages in these counties rose over the same period, increasing the value of travel time. Once the value of time is allowed to vary with median wages, the implied optimal tax and deadweight loss do not necessarily fall and can increase in some counties and periods, even as congestion declines. A decomposition of the change in the optimal tax shows that the wage-driven component is often dominant, exceeding the congestion-driven component by more than a factor of three in some cases.

To move beyond a simple before–after comparison and recover the demand side, I estimate how freeway travel demand responds to changes in remote work. A key concern is that changes in remote work and congestion may be jointly driven by unobserved local shocks. To address this, I complement the descriptive analysis with an instrumental variables strategy based on a shift–share design targeted at the demand side. I construct an instrument for the county-level share of people who work from home using occupation-level teleworkability indices from [Dingel and Neiman \(2020\)](#) interacted with pre-pandemic occupational composition. This shift–share measure provides plausibly exogenous variation in local work-from-home intensity that shifts the demand for freeway travel and allows me to identify the demand response to changes in remote work, separate from county-specific shocks.

These findings highlight that conclusions about welfare gains from reduced congestion depend critically on how the value of time and travel demand are modeled. A common approach in empirical congestion-pricing applications treats the value of time as fixed (or evolving only mechanically) and exogenous to policy, and often uses observed traffic volumes as a proxy for underlying demand conditions rather than explicitly modeling demand. In contrast, my results show that concurrent changes in wages and remote work can overturn the mechanical prediction that lower congestion implies lower optimal taxes and smaller welfare losses when the value of time is held constant. While this paper operationalizes the value of time using the median wage in each county and year, the results point to a natural extension in which the relevant value of time is that of the marginal driver and is allowed to depend on

the congestion tax itself through compositional changes in who chooses to drive.

This paper contributes to several strands of literature. A large body of work has examined how congestion responds to infrastructure and policy. [Duranton and Turner \(2011\)](#) show that expanding road capacity induces additional travel demand in U.S. cities, offsetting potential gains in congestion. A related literature studies road pricing instruments such as tolls and fuel taxes, arguing that congestion pricing is a more efficient tool for managing traffic ([Parry & Small, 2005](#); [Vandyck & Rutherford, 2018](#); [Verhoef, 2002](#)). Other studies link regional variation in average speed to economic outcomes and welfare, including [Couture, Duranton, and Turner \(2018\)](#) for U.S. cities and [Akbar, Couture, Duranton, and Storeygard \(2023\)](#) for Indian cities. Finally, a growing literature documents how the pandemic altered mobility patterns, modal choices, and externalities, including changes in traffic volumes, air quality, and road safety (e.g. [Andara et al., 2021](#); [Gu & Liu, 2023](#); [Gui, Wang, & Yu, 2022](#); [Li, Xu, & Li, 2021](#); [Xu, Li, Hu, Wu, & Li, 2022](#)).

To the best of my knowledge, this paper is the first to use the pandemic-era collapse in commuting demand to trace out changes in marginal external congestion costs and implied optimal congestion taxes, while explicitly incorporating time variation in the value of time. On the demand side, I estimate travel demand using a shift–share instrument for remote work exposure based on [Dingel and Neiman \(2020\)](#). More broadly, the results show that once the value of time is allowed to change with wages and remote work, lower congestion does not mechanically imply lower deadweight loss or lower optimal taxes.

The rest of the paper is organized as follows. Section 2 outlines the externalities framework, the demand-side specification, and the construction of MECC, optimal congestion taxes, and deadweight loss. Section 3 describes the data sources and key variables. Section 4 presents the empirical estimation of the supply and demand sides. Section 5 reports the main results. Section 6 concludes and discusses directions for future research.

2 Model

Building on the frameworks of [Quinet and Vickerman \(2004\)](#) and [Newbery \(1989\)](#), the marginal social cost of congestion is defined as

$$MSC(F) = \frac{\partial TSC}{\partial F} = ASC(F) + MECC(F), \quad (1)$$

where MSC denotes marginal social cost, $ASC(F)$ is the average social cost of congestion faced by each driver, and $MECC(F)$ is the marginal external congestion cost. The ASC represents the cost that users internalize when making travel decisions, while $MECC$ captures the additional time cost each driver imposes on others. Following [Yang et al. \(2020\)](#), the marginal external cost of congestion is given by

$$MECC(F) = F \cdot VOT \cdot \frac{\partial T}{\partial F}. \quad (2)$$

The $MECC$ framework provides a foundation to evaluate both the welfare implications of congestion and the corresponding adjustment in the optimal congestion price (Pigouvian tax). To operationalize Equations (1) and (2), I specify the functional form of $T(F)$ using a Bureau of Public Roads (BPR) function.

2.1 Travel Time and Value of Time

I model the relationship between traffic flow and delay using a BPR-type equation:

$$TTI_{c,h,t} = 1 + \alpha (F_{\text{obs},c,h,t})^\beta, \quad (3)$$

where $TTI_{c,h,t}$ denotes the Travel Time Index in county c , hour h , and year t . The term $TTI_{c,h,t} - 1$ represents the proportional delay relative to free-flow conditions (for example,

$TTI = 1.2$ implies 20% longer travel time).

Observed traffic flow $F_{\text{obs},c,h,t}$ is measured as vehicle miles traveled (VMT) per hour and defined as

$$F_{\text{obs},c,h,t} = VMT_{\text{Passenger},c,h,t} + \delta \cdot VMT_{\text{Truck},c,h,t}, \quad (4)$$

where δ is a truck-to-car equivalence factor reflecting the higher congestion contribution of trucks. Following FHWA guidelines, I set $\delta = 1.5$.²

Travel time per mile is given by $1/S_{\text{free-flow}}$, where $S_{\text{free-flow}}$ denotes the free-flow speed (60 mph in the PeMS data). Therefore, total travel time is

$$T_{c,h,t} = T_{\text{free-flow},c,t} \cdot TTI_{c,h,t}. \quad (5)$$

2.2 Computing Costs

The value of time (VOT) is defined as a constant fraction of the local wage:

$$VOT_{c,t} = \theta \cdot w_c, \quad (6)$$

where $\theta = 0.5$ following [Small \(2012\)](#). This implies that time is valued at half the median hourly wage in county c .

The derivative of travel time with respect to flow is

$$\frac{dT_{c,h,t}}{dF} = \frac{1}{S_{\text{free-flow}}} \cdot \alpha \cdot \beta \cdot (F_{\text{obs},c,h,t})^{\beta-1}. \quad (7)$$

²Typical values range from 1.5 to 2 depending on roadway type.

Substituting Equations (3), (7), and the definition of VOT into Equation (1) yields

$$MSC_{c,h,t} = \underbrace{\frac{VOT_{c,t}}{S_{\text{free-flow}}} \left[1 + \alpha (F_{\text{obs},c,h,t})^\beta \right]}_{\text{Average Social Cost (ASC)}} + \underbrace{\frac{VOT_{c,t}}{S_{\text{free-flow}}} \alpha \beta (F_{\text{obs},c,h,t})^\beta}_{\text{Marginal External Congestion Cost (MECC)}}. \quad (8)$$

The parameters α and β are estimated from Equation (3) using OLS.

2.3 Travel Decisions and Demand Specification

Drivers base their decisions on the average social cost $ASC(F_{c,t})$, which reflects the generalized price of travel. The equilibrium flow thus satisfies $P(F_{c,t}) = ASC(F_{c,t})$. Travel demand is expressed as

$$F_{c,t} = A_{ct} \cdot P_{ct}^{-\eta}, \quad (9)$$

where η is the price elasticity of travel demand, typically between -0.1 and -0.3 for VMT with respect to fuel costs. The demand shifter A_{ct} captures county-specific and temporal variations:

$$A_{ct} = \mu_c + \sigma Z_{ct} + X_{ct}\beta, \quad (10)$$

where μ_c are county fixed effects, X_{ct} are observable controls (for example, income, fuel price, and population), and Z_{ct} is the share of people working from home.

To address potential endogeneity of Z_{ct} , I first define

$$Z_{ct} \equiv \text{WFH share}_{ct}, \quad (11)$$

the share of employed residents in county c and year t whose usual place of work is their home. I then instrument Z_{ct} using a Bartik-type (shift–share) variable,

$$Z_{ct}^{IV} = \sum_j s_{cj}^{2019} \tau_{jt}^{WFH}, \quad (12)$$

where s_{cj}^{2019} is the 2019 employment share of industry j in county c , and τ_{jt}^{WFH} is the national change in work-from-home propensity in industry j at time t , constructed using teleworkability indices from [Dingel and Neiman \(2020\)](#).

Intuitively, Z_{ct}^{IV} captures how the work-from-home share in county c would evolve if its pre-pandemic industry structure were exposed only to national shocks to telework feasibility. The identifying assumption is that, conditional on controls and fixed effects, these national industry-level shocks are orthogonal to county-specific unobservables that directly affect congestion, so that Z_{ct}^{IV} shifts freeway travel demand only through its effect on remote work.

2.4 Freeway Optimal Congestion Tax

In equilibrium, $P(F_{c,t}) = ASC(F_{c,t})$. In the social optimum, the flow $F_{c,t}^{opt}$ satisfies

$$P(F_{c,t}^{opt}) = MSC(F_{c,t}^{opt}). \quad (13)$$

Given the demand in Equation (9), the optimal flow can be written as

$$F_{c,t}^{opt} = A_{ct} \left[\frac{1}{MSC(F_{c,t}^{opt})} \right]^\eta, \quad (14)$$

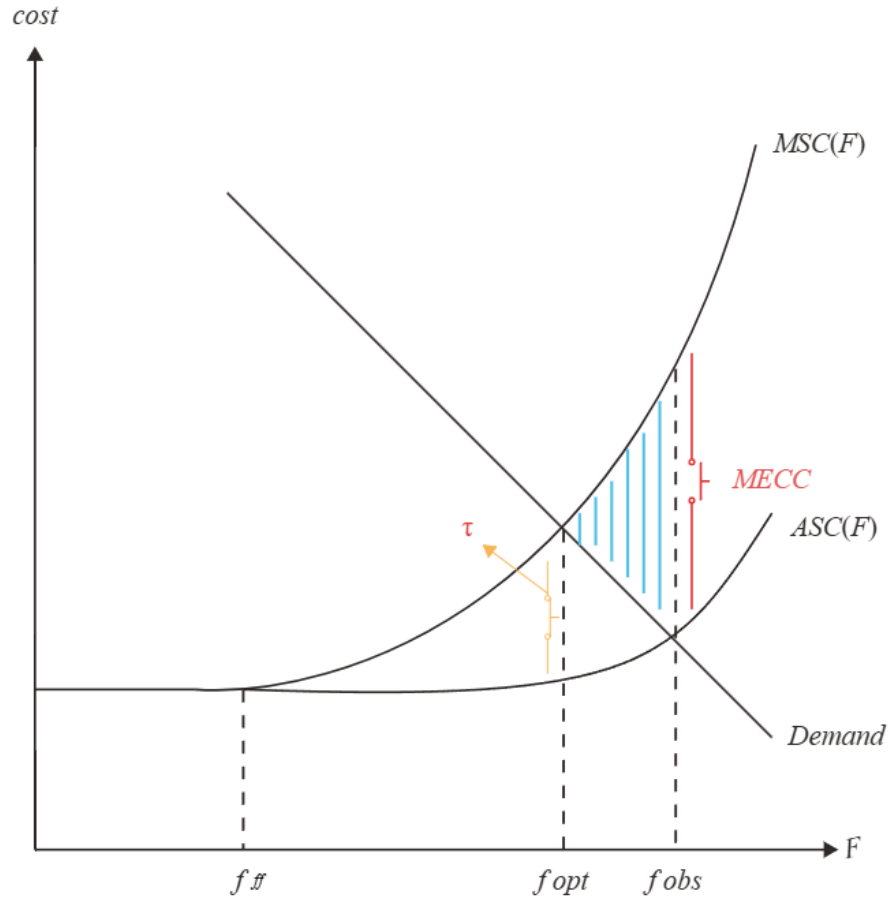
and the optimal Pigouvian tax is

$$\tau_{c,t} = MECC(F_{c,t}^{opt}). \quad (15)$$

Under a local linearity assumption for the marginal external congestion cost between $F_{c,t}^{opt}$ and $F_{obs,c,t}$, the deadweight loss (DWL) associated with unpriced congestion can be expressed as

$$DWL_{c,t} = \frac{1}{2} \cdot MECC(F_{obs,c,t}) \cdot (F_{obs,c,t} - F_{c,t}^{opt}). \quad (16)$$

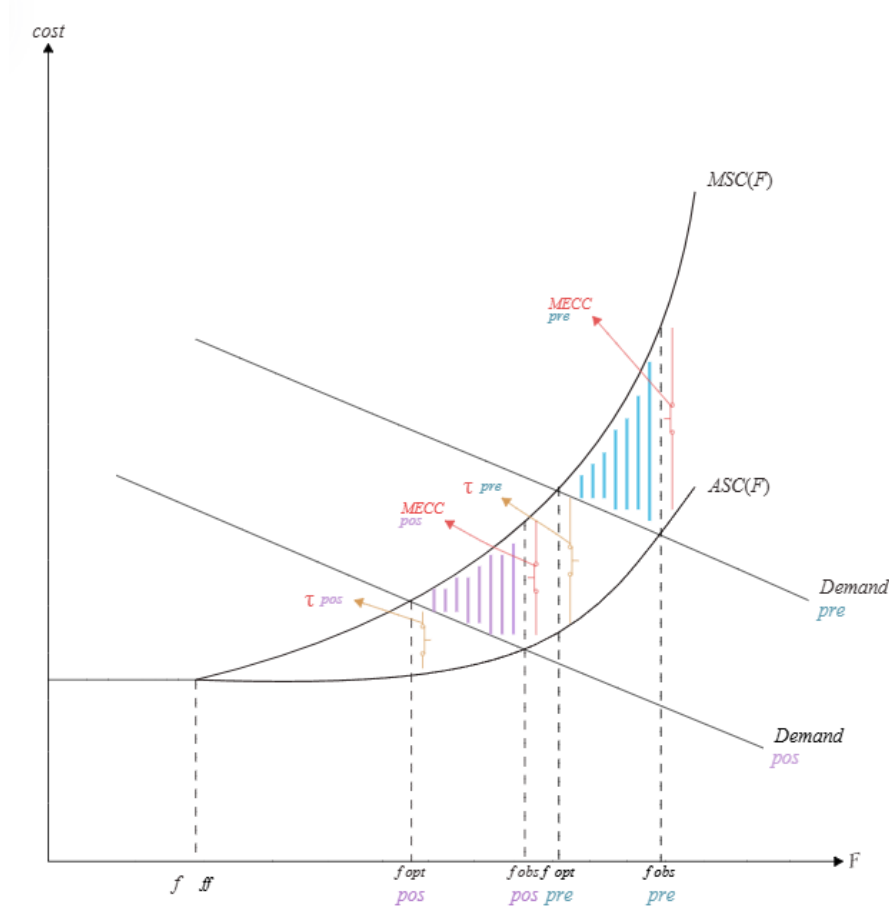
Figure 1. Average and Marginal Social Costs of Congestion



Notes: The Average Social Cost (ASC) represents the cost perceived by drivers, while the Marginal Social Cost (MSC) includes the external cost component (MECC). The shaded area represents the deadweight loss from unpriced congestion.

Following the pandemic, a negative demand shock shifts the travel demand curve downward, reducing observed flow and congestion. Holding VOT constant, changes in $MECC$ are driven entirely by this demand shift. The pandemic thus provides a natural experiment to identify the change in the optimal congestion tax, τ .

Figure 2. Optimal Tax Diagram with ASC, MECC, and MSC



Notes: This figure compares the pre- and post-pandemic equilibria. The difference between $\tau_{pre} = MECC(F_{opt,pre})$ and $\tau_{post} = MECC(F_{opt,post})$ reflects how optimal congestion pricing responds to the remote work-induced demand shock.

Finally, I compute the change in the optimal congestion tax as $\Delta\tau_c = \tau_{pre} - \tau_{post}$, and the corresponding welfare change as the difference in total deadweight loss across periods:

$$\Delta W_c = \sum_{t \in pre} DWL_{c,t} - \sum_{t \in post} DWL_{c,t}. \quad (17)$$

This framework links observed congestion data with optimal pricing and welfare, highlighting how shifts in travel demand, particularly through remote work, affect the efficiency of freeway usage.

2.5 Wage-Induced Shocks and the Value of Time

Finally, consider how changes in wages affect the equilibrium and optimal conditions in this framework. From (6), an increase in the local wage w_c raises the value of time $VOT_{c,t}$, which directly scales both the average social cost (ASC) and the marginal social cost (MSC) in Equation (8). Taking the partial derivative of the MSC with respect to the wage yields

$$\frac{\partial MSC_{c,h,t}}{\partial w_c} = \frac{\theta}{S_{\text{free-flow}}} \left[1 + \alpha(1 + \beta) (F_{\text{obs},c,h,t})^\beta \right] > 0. \quad (18)$$

Hence, higher wages increase the monetary valuation of travel time losses, shifting the entire cost curve upward. This represents a positive supply shock: at any given flow level, the perceived cost of travel is higher. As a result, the equilibrium flow F_{obs} decreases, since the intersection with the downward-sloping demand curve occurs at a lower quantity and higher generalized cost.

Implications for Optimal Flow and Congestion Pricing The same wage shock also affects the socially optimal flow $F_{c,t}^{\text{opt}}$ and the optimal tax $\tau_{c,t}$. Differentiating the optimality condition $P(F_{c,t}^{\text{opt}}) = MSC(F_{c,t}^{\text{opt}})$ with respect to w_c gives

$$\frac{\partial F_{c,t}^{\text{opt}}}{\partial w_c} = - \frac{\frac{\partial MSC}{\partial w_c}}{\frac{\partial MSC}{\partial F} - \frac{\partial P}{\partial F}} < 0, \quad (19)$$

implying that optimal flow decreases as wages rise. Because the marginal external cost $MECC(F)$ is proportional to VOT , the optimal congestion tax increases with the value of time:

$$\frac{\partial \tau_{c,t}}{\partial w_c} = \frac{\partial MECC(F_{c,t}^{\text{opt}})}{\partial w_c} = \frac{\theta}{S_{\text{free-flow}}} \alpha \beta (F_{c,t}^{\text{opt}})^\beta > 0. \quad (20)$$

Therefore, higher-income regions not only experience higher perceived travel costs but also require higher optimal congestion charges to internalize the corresponding increase in

the opportunity cost of time. Spatial heterogeneity in wages thus translates directly into spatial heterogeneity in both the efficient level of travel and the optimal congestion pricing policy.

2.6 Decomposition of Changes in the Optimal Congestion Tax

In order to identify to which part contributes the most to the change in the tax in our data, the optimal Pigouvian congestion tax per mile at time t be written as

$$\tau_t = VOT_t g_t, \quad (21)$$

where VOT_t is the value of travel time (dollars per hour) and g_t denotes the marginal external travel delay per mile (hours per mile) implied by the congestion technology evaluated at optimal traffic conditions. In the model underlying equation (8), g_t is given by:

$$g_t = \frac{1}{S_{\text{free-flow}}} \alpha \beta (F_t^{\text{opt}})^\beta \quad (22)$$

Two-period change. Consider two periods, a pre-period and a post-period. Define

$$\Delta\tau \equiv \tau_{\text{post}} - \tau_{\text{pre}}. \quad (23)$$

Exact decomposition (counterfactual approach). Define the counterfactual tax that holds VOT at its pre-period level while allowing congestion to change:

$$\tau^g \equiv VOT_{\text{pre}} g_{\text{post}}. \quad (24)$$

Then,

$$\begin{aligned}\Delta\tau &= \tau_{\text{post}} - \tau_{\text{pre}} \\ &= \underbrace{(\tau^g - \tau_{\text{pre}})}_{\Delta\tau_{\text{cong}}} + \underbrace{(\tau_{\text{post}} - \tau^g)}_{\Delta\tau_{\text{vot}}}.\end{aligned}\tag{25}$$

Substituting from (21) and (24) yields

$$\Delta\tau_{\text{cong}} = VOT_{\text{pre}} (g_{\text{post}} - g_{\text{pre}}),\tag{26}$$

$$\Delta\tau_{\text{vot}} = g_{\text{post}} (VOT_{\text{post}} - VOT_{\text{pre}}).\tag{27}$$

Equations (26)–(27) provide an exact decomposition:

$$\Delta\tau = \Delta\tau_{\text{cong}} + \Delta\tau_{\text{vot}}.\tag{28}$$

3 Data

This section introduces the data used in the analysis and documents how the COVID-19 pandemic produced a large, sudden, and persistent reduction in congestion across California’s main metropolitan counties. The observed patterns are consistent with a negative demand shock to travel behavior that reduced vehicle flows and travel times.

3.1 Data Sources

The main dataset used in this study is the California Performance Measurement System (PeMS), a real-time database managed by Caltrans that records hourly traffic conditions from more than thirty-nine thousand loop detectors across the state. PeMS provides information on Vehicle Miles Traveled (VMT), Vehicle Hours Traveled (VHT), average speed, and the

Travel Time Index (TTI), which measures the ratio of actual travel time to free-flow travel time. In PeMS, the free-flow speed is fixed at sixty miles per hour, so TTI values above one indicate congestion.

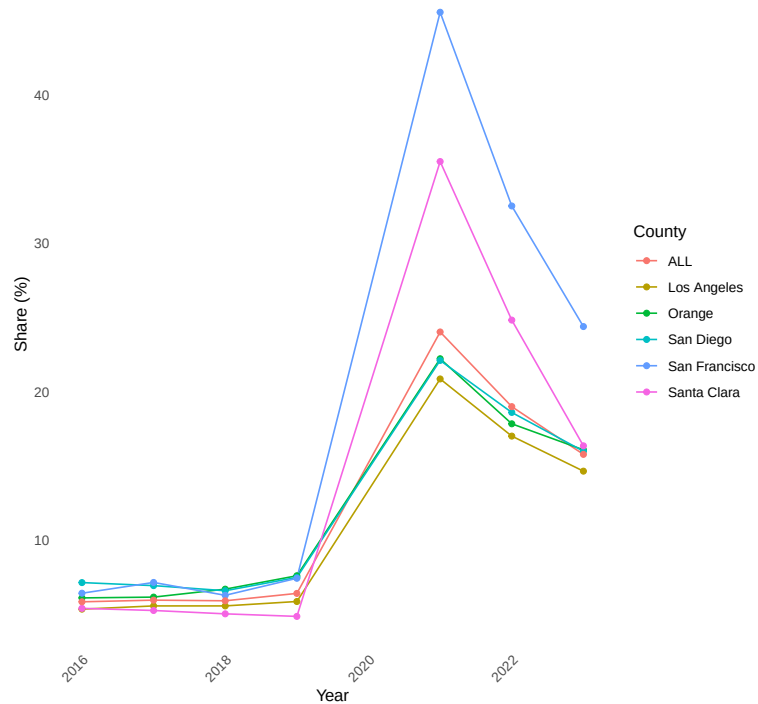
To complement PeMS, I use data from the American Community Survey (ACS) to obtain median wages, population, and the share of people working from home. These variables are later used to compute the value of time (VOT) and to interpret behavioral changes in travel patterns.

3.2 The Pandemic as a Natural Experiment

The COVID-19 pandemic created a unique setting to study congestion and welfare. In March 2020, California implemented a statewide lockdown that sharply reduced mobility across all urban areas. Because no major freeway expansions, closures, or pricing policies occurred during this period, the pandemic can be interpreted as an exogenous event that altered travel behavior through restrictions on movement and the expansion of remote work, rather than through changes in road supply.

Figure 3 shows the evolution of the share of people working from home from 2016 to 2023. Before the pandemic, remote work was rare, remaining below seven percent in all counties. In 2021, it rose sharply, exceeding forty-five percent in San Francisco and about thirty-five percent in Santa Clara. Although the share of remote workers declined slightly after 2022, it stabilized at a much higher level than before the pandemic. This sustained increase in remote work represents the main behavioral channel through which the pandemic reduced travel demand.

Figure 3. Share of People Working from Home by County

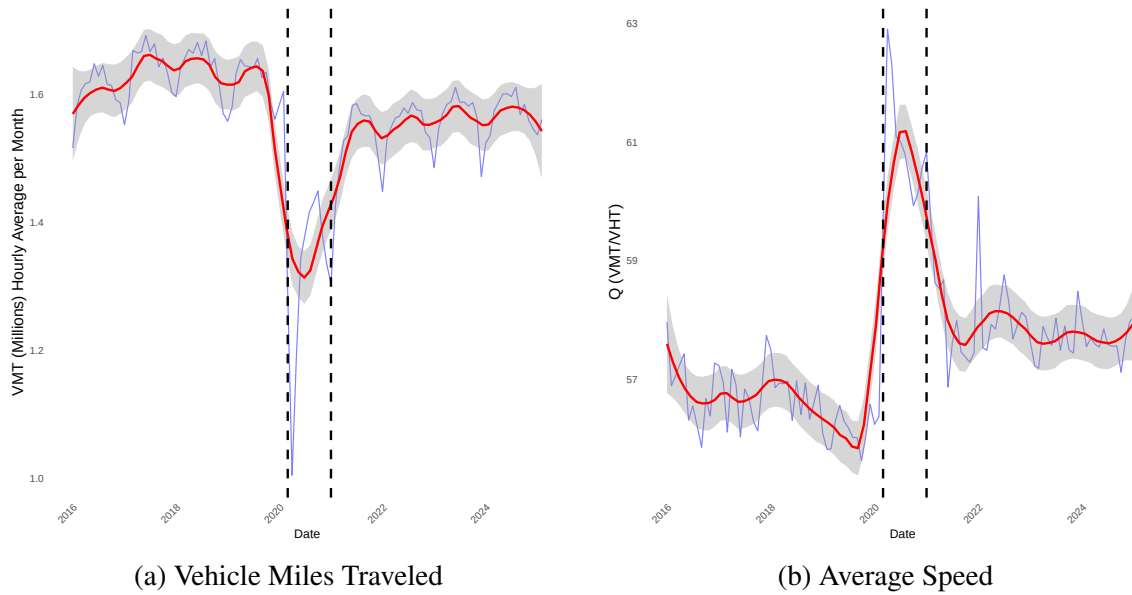


Notes: Data from the American Community Survey. The figure shows a sharp and persistent rise in remote work after 2020, especially in San Francisco and Santa Clara.

3.3 Changes in Traffic and Congestion

Traffic data reveal a clear and lasting change in freeway conditions following the pandemic. Figure 4 shows that Vehicle Miles Traveled (VMT) declined sharply during the lockdown and never returned to its previous level. Average speed increased simultaneously, indicating a reduction in congestion. The pattern suggests a structural, not temporary, shift in travel behavior.

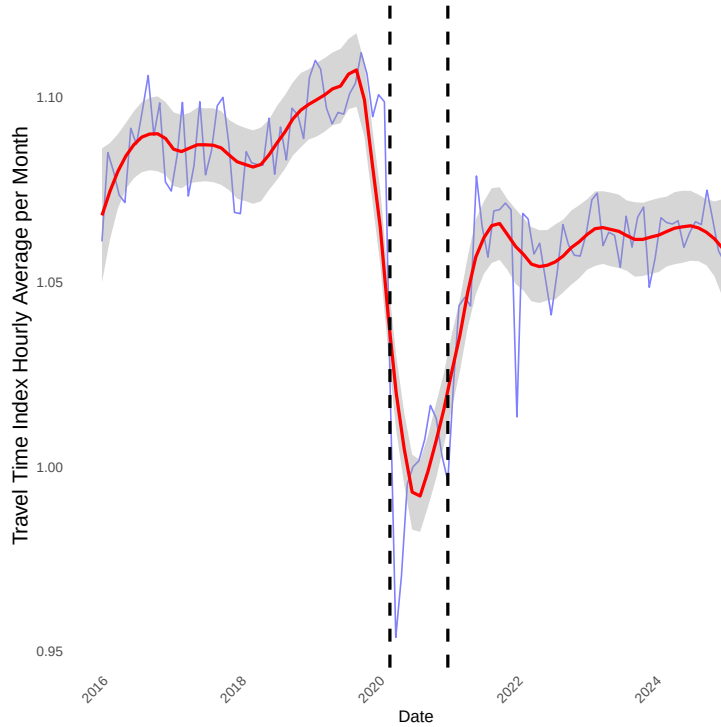
Figure 4. Vehicle Miles Traveled and Average Speed



Notes: Monthly county averages from PeMS for the years 2016 to 2025. The red line shows the smoothed trend, highlighting a persistent decline in VMT and a sustained increase in average speeds after 2020.

The Travel Time Index displayed in Figure 5 confirms the same pattern. After 2020, TTI values dropped sharply, showing that travel times became shorter relative to free-flow conditions. This evidence indicates that congestion declined substantially in the post-pandemic period across all counties in the sample.

Figure 5. Travel Time Index



Notes: Monthly averages from PeMS for the years 2016 to 2025. Lower TTI values correspond to shorter travel times and less congestion.

Table 1 presents a before-and-after comparison for VMT and average speed using hourly data for the years 2018 to 2019 and 2021 to 2022. The results show a significant decline in VMT and an increase in speed, confirming the descriptive evidence. The changes are strongest during rush-hour periods, when congestion had previously been most severe.

Figures 6 and 7 further illustrate how the effects vary by hour of the day. The largest improvements in travel time and speed occurred during the morning and afternoon peaks. Even modest reductions in traffic volume at these times produced large gains in speed, consistent with the nonlinear nature of congestion.

Table 1: Differences in Traffic Before and After the Pandemic

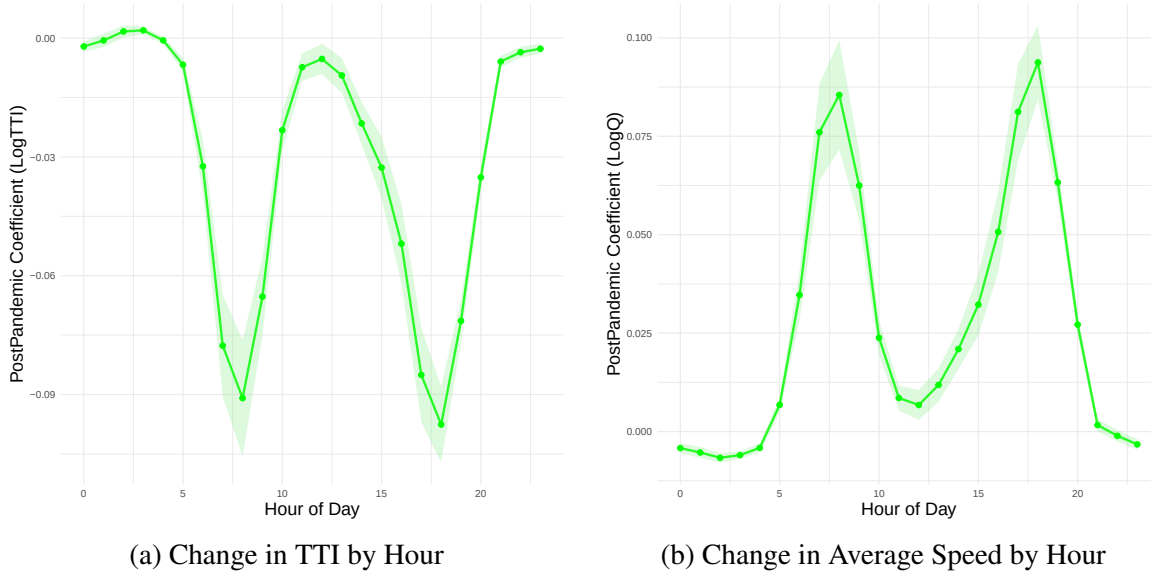
	VMT All Data	Q	VMT (3pm6pm)	Q	VMT Weekends	Q
	(1)	(2)	(3)	(4)	(5)	(6)
Pandemic After	−0.067*** (0.006)	0.028*** (0.001)	−0.040*** (0.013)	0.066*** (0.002)	−0.054*** (0.012)	0.006*** (0.001)
Constant	13.642*** (0.004)	4.080*** (0.0004)	14.200*** (0.009)	3.902*** (0.001)	13.569*** (0.008)	4.134*** (0.001)
Observations	181,655	181,655	30,280	30,280	51,695	51,695
R ²	0.001	0.011	0.0003	0.051	0.0004	0.001
Adjusted R ²	0.001	0.011	0.0003	0.051	0.0004	0.001

Note:

*p<0.1; **p<0.05; ***p<0.01

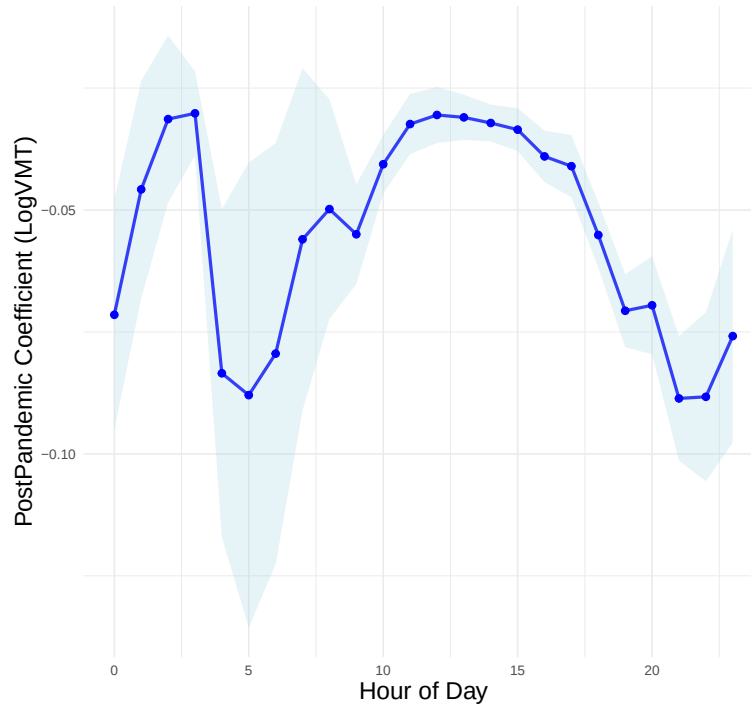
Note: In each section, the first two columns examine the difference in means of $\log(Q)$ and $\log(VMT)$ before and after the pandemic. Columns 1 and 2 include all hourly data from 2018 to 2022, columns 3 and 4 focus on rush hour data (3 PM to 6 PM), and columns 5 and 6 analyze weekend data.

Figure 6. Changes in Congestion by Hour



Notes: Hourly regression estimates comparing 2018–2019 to 2021–2022. The most pronounced improvements occurred during peak commuting hours.

Figure 7. Changes in Vehicle Miles Traveled by Hour

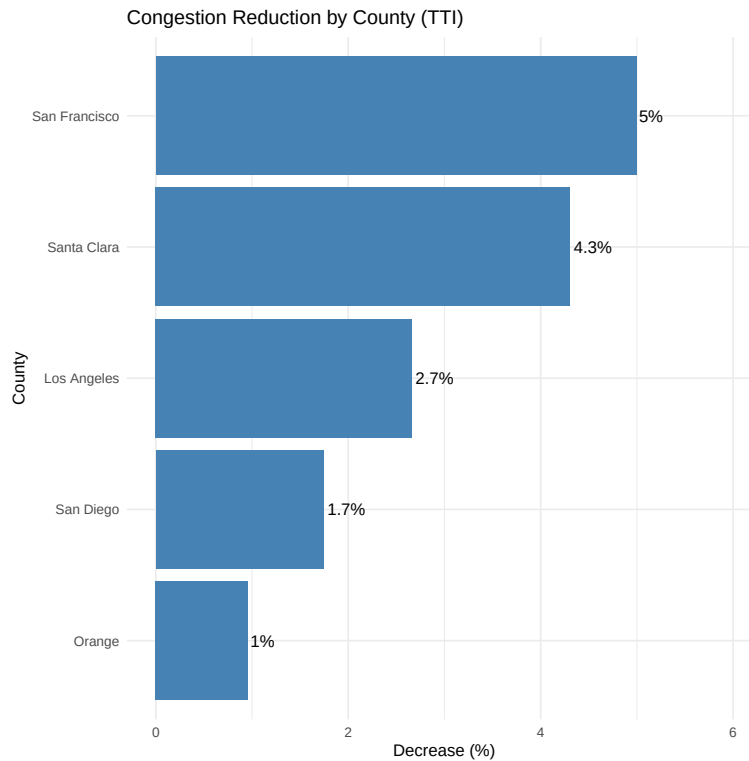


Notes: Hourly regression estimates comparing 2018–2019 to 2021–2022. VMT declined most sharply during early morning hours and remained relatively stable at night.

3.4 Differences Across Counties

The reduction in congestion was not uniform across counties. Figure 8 shows that San Francisco and Santa Clara experienced the largest improvements, followed by Los Angeles, San Diego, and Orange. These differences closely match the intensity of remote work adoption, with counties that had the highest increase in work-from-home rates also showing the greatest decline in congestion.

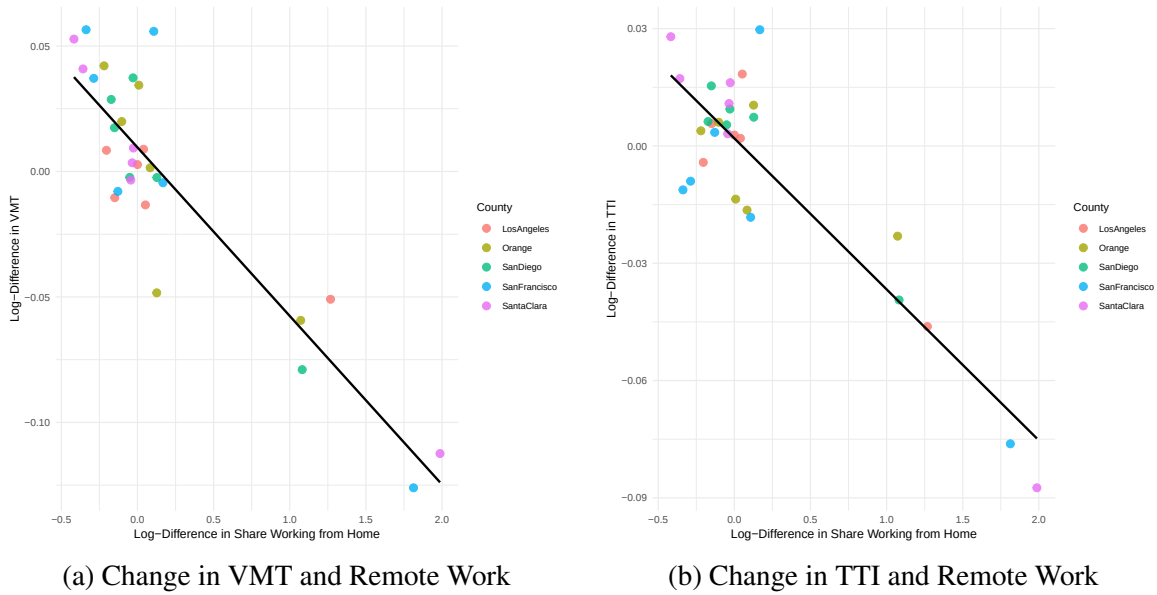
Figure 8. Change in Congestion by County



Notes: Estimated changes in the Travel Time Index by county. Counties with larger increases in remote work saw greater reductions in congestion.

The relationship between remote work and traffic is further illustrated in Figure 9, which compares the yearly change in remote work shares with changes in VMT and TTI. Counties that experienced the largest increases in remote work also recorded the largest declines in traffic and congestion. This pattern supports the interpretation that the pandemic primarily affected congestion through a reduction in travel demand.

Figure 9. Changes in Traffic and Congestion versus Remote Work



Notes: Each point represents a county-year observation. Data from PeMS and the American Community Survey. Counties with larger increases in remote work show stronger declines in VMT and TTI.

Alternative explanations for the decline in congestion are less consistent with the data. Freight activity, measured by truck VMT, accounts for less than five percent of total VMT and declined only slightly after the pandemic. Figure A.1 shows the reduction in VMT separately for trucks and non-truck vehicles, with patterns that closely mirror the aggregate results presented above. Population levels fell modestly between 2020 and 2021 but recovered in subsequent years, and average commuting distances remained stable. Taken together, these facts point to a primarily behavioral rather than structural source of the observed congestion relief.

4 Estimation

4.1 Supply side

I start by estimating the BPR equation in (3) using OLS, as specified below:

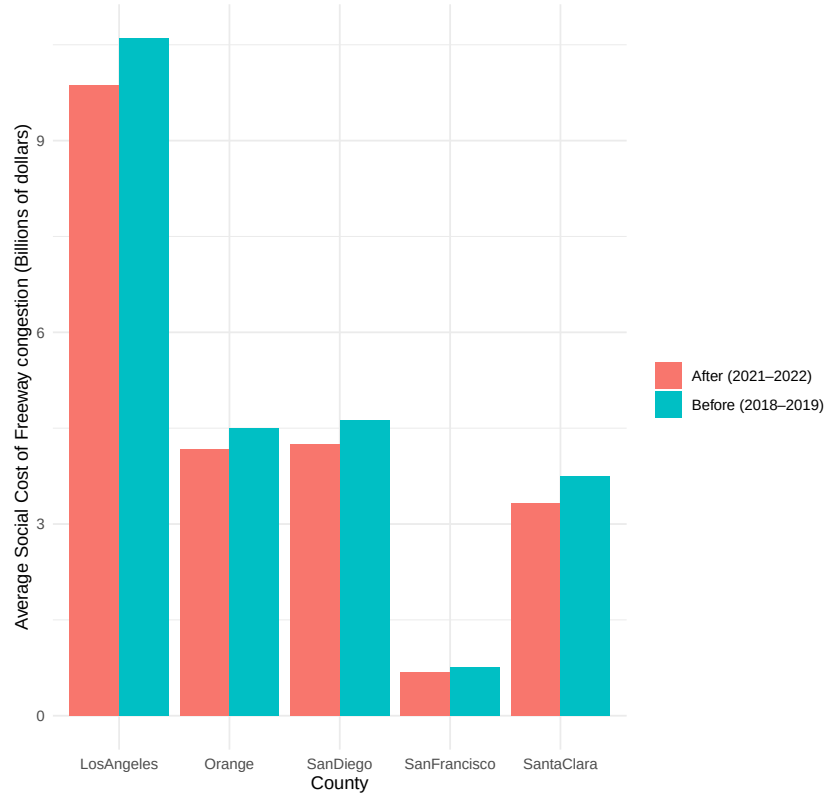
$$\ln(TTI_{c,h,t} - 1) = \rho + \beta \cdot \ln(F_{\text{obs},c,h,t}) + \gamma_c + \epsilon_{c,h,t}, \quad (29)$$

where γ_c denotes county fixed effects. These fixed effects absorb time-invariant cross-county differences in baseline travel conditions, including persistent differences in highway infrastructure, network design, and other structural features of the road system. Because major changes in aggregate freeway capacity are limited over the relatively short analysis window, treating these county-specific factors as stable is a reasonable identifying assumption.

The estimated congestion elasticity is $\beta = 3.6$, which lies comfortably within the range commonly used in the congestion-pricing literature (roughly 2.5 to 5) (Small, Verhoef, & Lindsey, 2024). Given this estimate, the BPR scale parameter is recovered as $\alpha = \exp(\rho)$.

As for the value of time, I use the constant proportion $\theta = 0.5$ of the median wage, as mentioned above. Given the coefficients in Equation (29), θ , and median wages in 2018 held fixed, I compute the social cost of congestion. Figure 10 shows the average total social cost of freeway congestion in each county before and after the pandemic.

Figure 10. Average Total Social Cost of Congestion



Notes: Average social cost of congestion before and after the pandemic for each county, using 2018 median wages to compute the value of time. The measure is based on Equation (8), multiplied by flow and averaged for each county before and after the pandemic.

The average social cost of congestion after the pandemic is lower in all counties than before the pandemic. It is important to note that this is freeway congestion. Therefore, it depends heavily on freeway use; since San Francisco relies less on freeways, its social cost measure is comparatively lower, while in LA County it is above 9 billion dollars.

4.2 Demand side

To estimate the demand for freeway travel, I use two-stage least squares (2SLS), instrumenting the potentially endogenous remote work variable Z_{ct} with the shift–share instrument defined in Equation (12).

Demand equation (second stage). The demand for freeway travel is specified as

$$\ln F_{ct} = \alpha_c - 0.1 \ln(\text{gas_price}_{ct}) + \beta_Z Z_{ct} + u_{ct}, \quad (30)$$

where α_c are county fixed effects, gas_price_{ct} is the local gasoline price, $Z_{ct} = \text{share_Home}_{ct}$ is the share of workers who work from home in county c and year t , and u_{ct} is an error term. The coefficient β_Z captures the causal effect of remote work on freeway demand. The coefficient on gasoline prices is imposed rather than estimated: I fix the elasticity of freeway traffic with respect to gasoline prices at -0.1 , consistent with the short-run price elasticities of Vehicle Miles Traveled reported in the transportation economics literature.³

First stage (2SLS). Because Z_{ct} may be endogenous, I estimate Equation (30) using 2SLS. In the first stage, the work-from-home share is projected on the shift-share instrument:

$$Z_{ct} = \delta_c + \pi Z_{ct}^{IV} + \eta_{ct}, \quad (31)$$

where δ_c are county fixed effects, Z_{ct}^{IV} is the Bartik (shift-share) instrument constructed from pre-pandemic industry shares (2018 shares) and national shocks to telework feasibility, and η_{ct} is the first-stage error term.

Substituting the fitted values $\widehat{Z}_{ct}$ from Equation (31) into Equation (30) yields the 2SLS estimating equation:

$$\ln F_{ct} = \alpha_c - 0.1 \ln(\text{gas_price}_{ct}) + \beta_Z \widehat{Z}_{ct} + \varepsilon_{ct}, \quad (32)$$

where ε_{ct} denotes the composite error term in the second stage.

³Short-run elasticities of driving with respect to fuel or gasoline prices are typically small in magnitude, often in the range of -0.05 to -0.25 , with values around -0.1 being a common benchmark in both empirical work and policy simulations.

The 2SLS results are summarized in Table 2.

Table 2: Effect of Working from Home on Freeway Flow (2SLS)

Dependent Variable:	log(F)
Model:	(1)
<i>Variables</i>	
WFH share (2SLS)	-0.0022*** (0.0002)
<i>Fixed-effects</i>	
county	Yes
<i>Fit statistics</i>	
F-test (1st stage), share _{Home}	29.344
Wu-Hausman, p-value	0.07386
Observations	20
R ²	0.99971
<i>Clustered (county) standard-errors in parentheses</i>	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

The two-stage least squares estimates reported in Table 2 show a strong negative effect of working from home on freeway traffic volumes. The coefficient on the instrumented work-from-home share is -0.0022 with a clustered standard error of 0.0002 , statistically significant at the 1 percent level. This implies that an increase in the share of workers who work from home is associated with a proportional decline in freeway flow. If the work-from-home share is measured in percentage points, a 10 percentage point increase in remote work reduces freeway traffic volumes by approximately 2.2 percent, holding constant gasoline prices and county fixed effects.

The fit statistics further support the validity of the 2SLS strategy. The first-stage F-statistic for the excluded instrument is 29.3, which comfortably exceeds standard weak-instrument thresholds and indicates that the shift-share instrument is strongly correlated with the endogenous work-from-home share. The Wu-Hausman p-value is 0.074, providing suggestive evidence that treating the work-from-home share as endogenous may matter, even though the

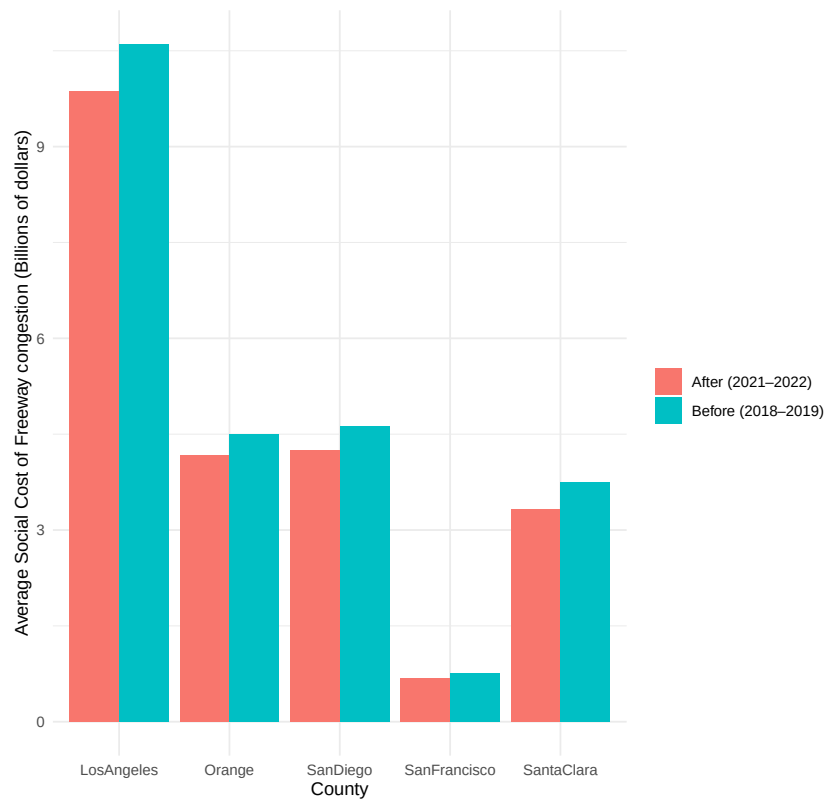
null of exogeneity is not rejected at the conventional 5 percent level. Overall, the 2SLS results are consistent with the interpretation that exogenous increases in remote work substantially depress freeway travel demand.

5 Results

5.1 Fixed Value of Time

Given the demand and supply estimates, I compute the optimal flow from Equation (14) and the deadweight loss from Equation (16). Figure 11 shows the average deadweight loss of congestion before and after the pandemic. The pattern is similar to that of the average social cost of congestion, although the magnitude is naturally smaller.

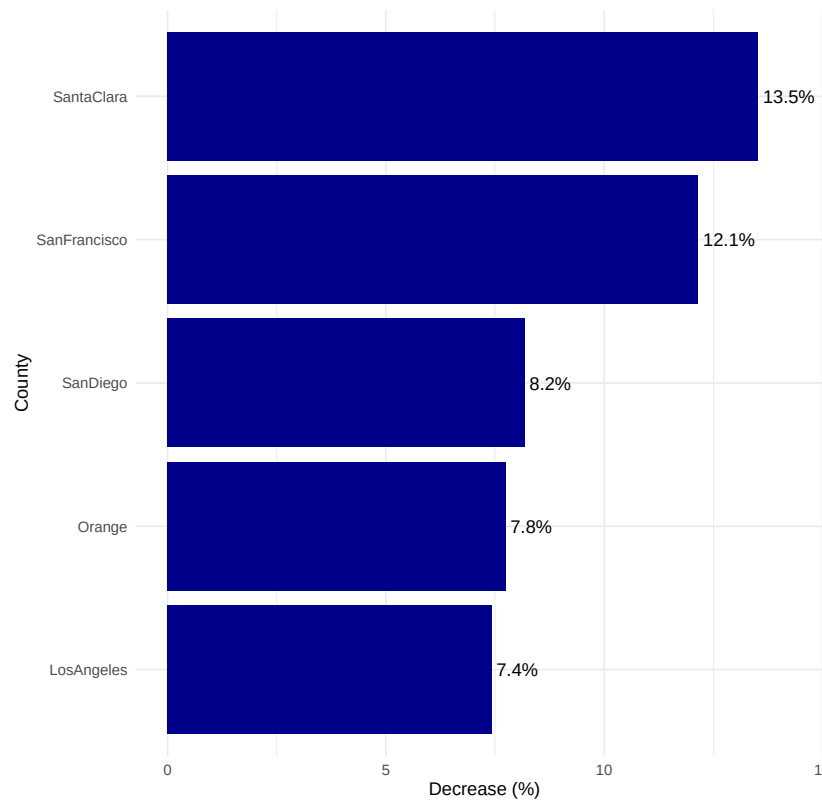
Figure 11. Deadweight Loss by County



Notes: Deadweight loss of congestion in terms of the value of time. This is a linear approximation using Equation (16) and keeping median wages from 2018 fixed.

Figure 12 presents the variation in deadweight loss between the pre- and post-pandemic periods. Santa Clara and San Francisco experienced the largest relative reductions in DWL, reflecting both high work-from-home shares and high values of time driven by median wages.

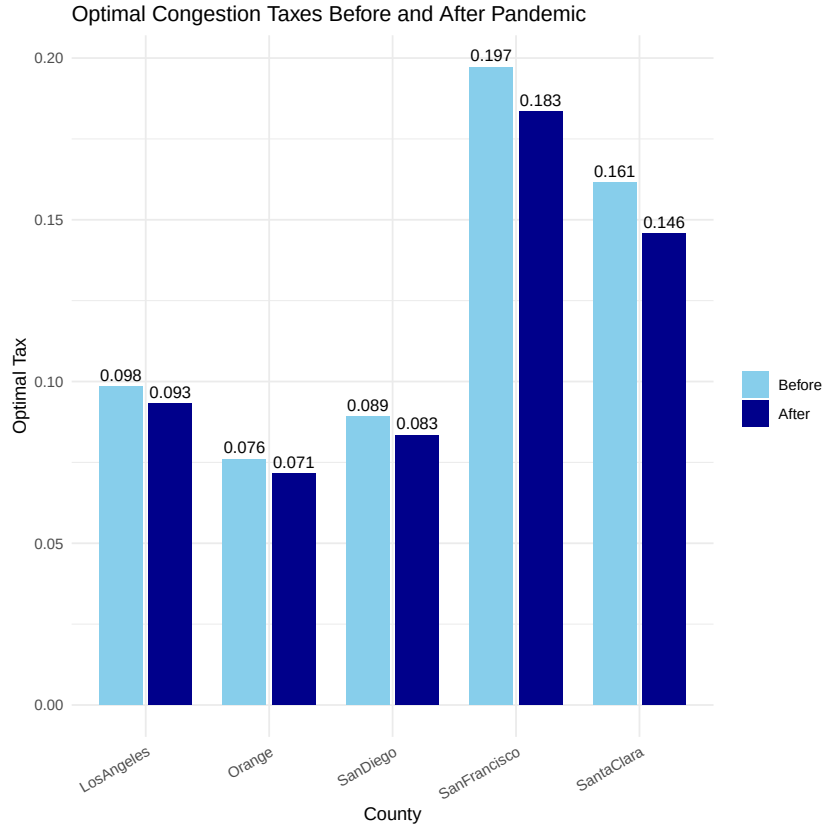
Figure 12. Deadweight Loss Variation by County



Notes: Variation in the deadweight loss of congestion across counties, using Equation (17) and keeping median wages from 2018 fixed.

In terms of optimal taxation, Figure 13 shows the optimal congestion tax per mile in cents before and after the pandemic in each county.

Figure 13. Optimal Congestion Tax per Mile (Fixed VOT)



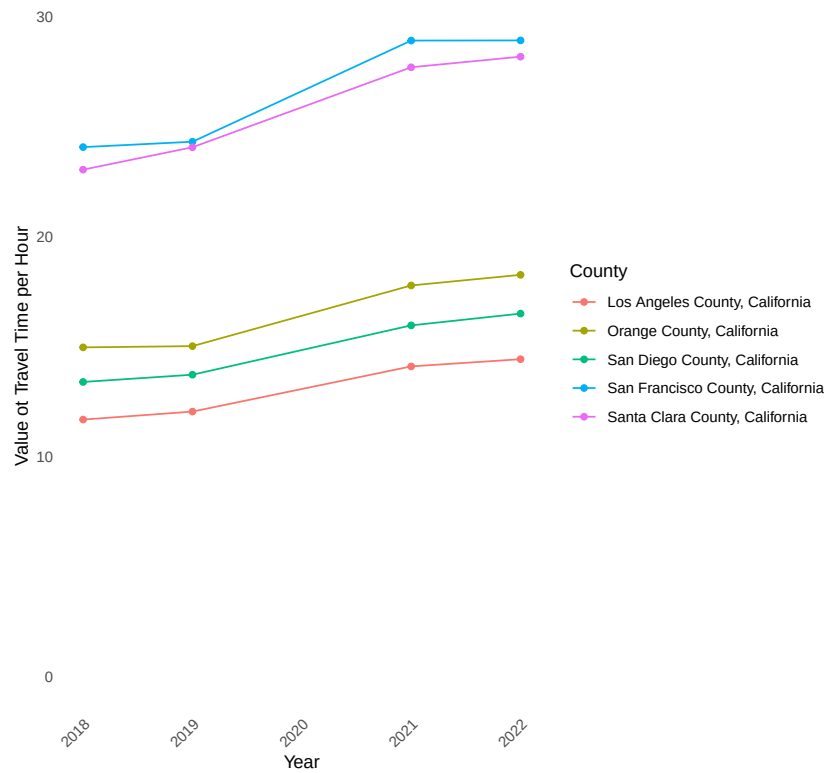
Notes: Optimal congestion tax per mile in cents. The tax is the marginal external cost of congestion evaluated at the optimal flow, as shown in Equation (15), holding the value of time fixed over time using 2018 median wages.

Even though the DWL is smaller in San Francisco and Santa Clara, the tax needed to eliminate it is higher because the value of time is higher in these counties. However, since there was a clear reduction in freeway congestion, the optimal taxes are lower after the pandemic when the value of time is held fixed.

5.2 Increasing Value of Time

The literature typically abstracts from changes in the value of time when studying the dynamics of optimal congestion taxes. If I do not hold wages fixed at a pre-pandemic level, the evolution of the value of time by county (recall that I use $VOT = 0.5 \times w_c$) is shown in Figure 14.

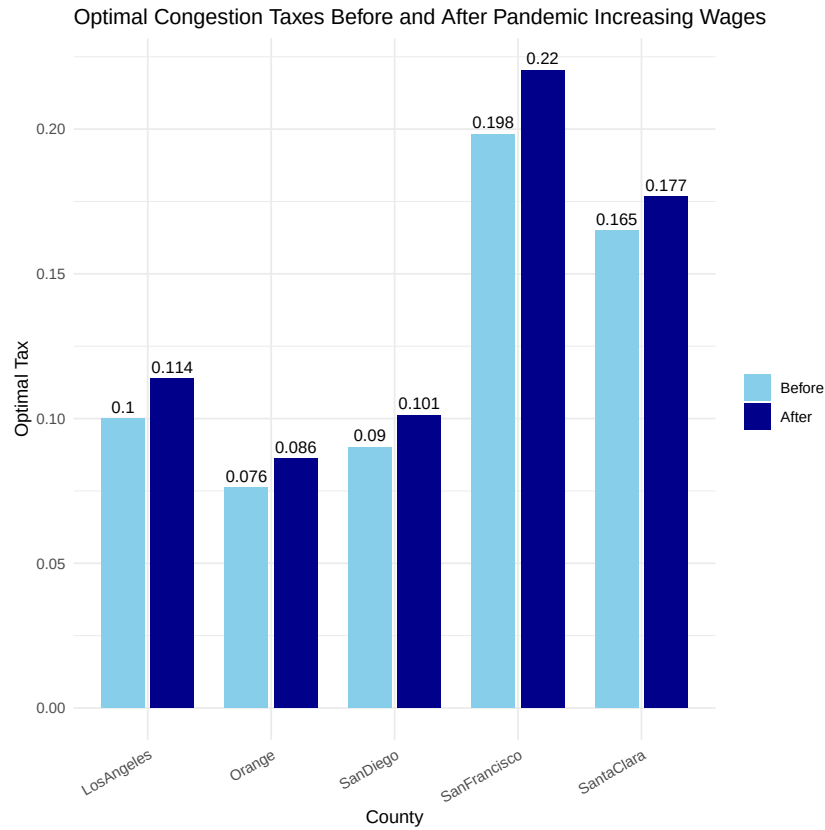
Figure 14. Value of Time by County



Notes: Value of time as a fixed proportion of the median wage in each county. The increase over time is particularly pronounced in San Diego and San Francisco.

This increase in the value of time translates into higher implied optimal taxes once wages are allowed to vary, as shown in Figure 15.

Figure 15. Optimal Taxes with Time-Varying Value of Time



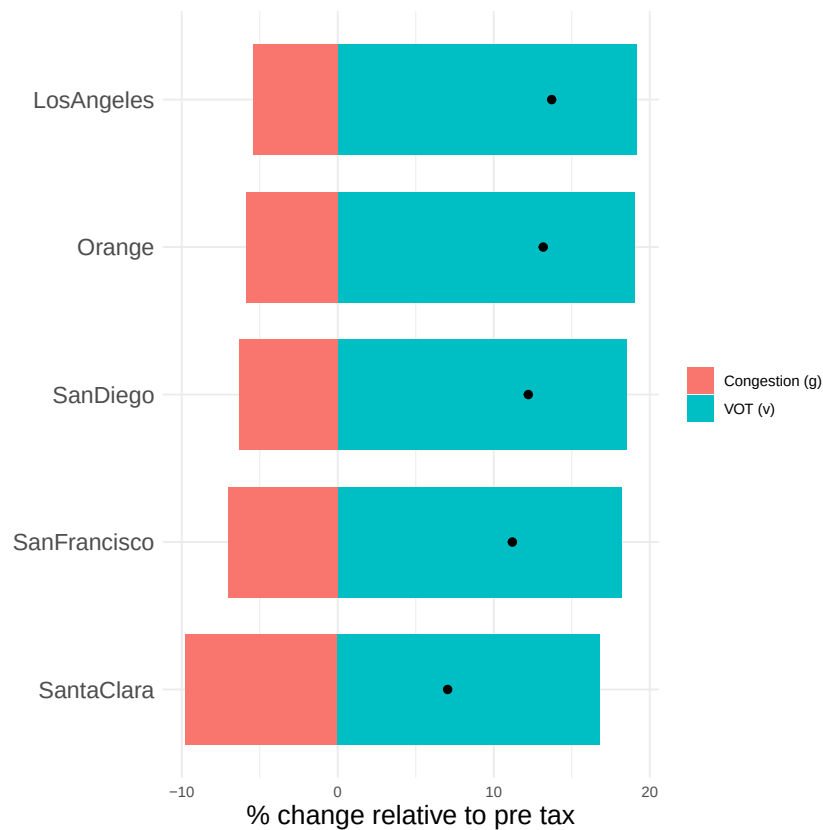
Notes: Implied optimal tax by county and year, allowing the value of time to vary with the median wage in each county.

Allowing wages to rise raises the implied optimal tax in each county, offsetting a substantial share of the mechanical decline that would otherwise follow from reduced congestion. The mechanism is straightforward: when time becomes more valuable, a given minute of delay imposes a larger external cost, so the Pigouvian tax need not fall one-for-one with improvements in traffic conditions. Throughout the paper, I proxy the value of time using the median wage in each county and year. A way to improve this in future research is to allow the relevant value of time to be that of the marginal driver, which may itself respond to congestion pricing if the tax changes the composition of who chooses to drive. If higher congestion taxes disproportionately deter lower-wage drivers, the marginal driver may have a higher wage than the median, implying that the optimal tax could be higher than the level

implied by a median-wage calibration.

Figure 16 decomposes the change in the implied optimal tax using equation (28). Across counties and years, the component attributable to changes in the value of time accounts for most of the variation and often dominates the component attributable to changes in congestion. Absent the increase in the value of time, the congestion component would predict lower optimal taxes (the orange portion of the figure), but this effect is largely offset once time-varying value of time is incorporated (green portion).

Figure 16. Decomposition of Changes in the Optimal Tax



Notes: The figure decomposes changes in the implied optimal tax into a congestion component and a value-of-time component as in equation (28).

6 Conclusion

This paper uses the COVID-19 pandemic as a large, plausibly exogenous shock to freeway travel demand to study how optimal congestion taxes and welfare respond to sustained changes in congestion and wages. Combining detailed PeMS data on freeway flows, speeds, and congestion with ACS measures of wages and work-from-home shares, I quantify changes in marginal external congestion costs, optimal Pigouvian taxes, and deadweight loss in five major California counties before (2018–2019) and after (2021–2022) the pandemic shock.

Three main findings emerge. First, freeway congestion fell sharply and persistently in the post-pandemic period, especially during peak hours, and this reduction in congestion is closely tied to the rise in remote work. A two-stage least squares demand specification, using a shift-share instrument for remote work based on [Dingel and Neiman \(2020\)](#), shows that increases in working from home have a sizeable negative effect on freeway travel demand: a 10 percentage point increase in the work-from-home share reduces freeway traffic by roughly 2.2 percent. This confirms that remote work is an important demand-side driver of congestion relief.

Second, when the value of time is held fixed at pre-pandemic wage levels, lower congestion translates into lower marginal external costs, smaller deadweight losses, and lower optimal congestion taxes in all counties. Under this conventional assumption, the pandemic-induced demand shock looks like a free, one-time congestion “policy”: welfare improves, and the Pigouvian tax needed to internalize congestion falls.

Third, once wages are allowed to rise, and with them the value of time, this conclusion no longer holds. Median wages increased substantially over the sample period, especially in higher-income counties. Because marginal external congestion costs are proportional to the value of time, higher wages shift the marginal cost curves upward and raise the implied optimal congestion tax, even when traffic volumes and measured congestion fall. In every county

in the sample, the wage-driven component dominates the congestion-driven component, as shown in the decomposition, so the implied optimal tax increases in the post-pandemic period. This result underscores that lower congestion does not mechanically translate into lower deadweight loss or lower optimal taxes once the value of time is allowed to vary.

These findings have several implications for policy and for the congestion pricing literature. From a policy perspective, the results suggest that calibrating congestion charges using outdated wage information or assuming a fixed value of time is likely to underprice congestion in growing, high-wage regions. Remote work may reduce traffic volumes, but it simultaneously raises the opportunity cost of time for many commuters, so that the welfare gain from an additional minute saved in traffic is larger. Accounting for both margins is essential for setting efficient congestion taxes. The results also highlight that remote work policies and congestion pricing are complements rather than substitutes: remote work reduces demand, but optimal pricing remains necessary to allocate scarce road space efficiently when the value of time rises.

From a conceptual standpoint, the paper points toward a richer modeling of the value of time in optimal congestion taxation. I operationalize the value of time using the median wage in each county and year, but the true relevant margin for optimal pricing is the wage of the marginal driver. If congestion charges affect the composition of drivers, for example by pushing lower-wage commuters onto alternative modes, then the wage of the marginal road user becomes a function of the tax itself. In that scenario, both the optimal tax and the value of time are endogenous and jointly determined. Incorporating this feedback, along with heterogeneous preferences for remote work and mode choice, is a promising direction for future research.

Finally, the analysis in this paper is deliberately focused on freeways in five California counties and abstracts from modal shifts, land use changes, and second-best interactions with other policies. Extending the framework to a multimodal network, allowing for transit

crowding, endogenous route choice, and environmental externalities, would provide a more complete welfare assessment of post-pandemic mobility patterns. Nonetheless, the central message is robust: in a world where travel demand, work arrangements, and wages are all evolving, optimal congestion taxes must adapt not only to how much people drive, but also to how valuable their time has become.

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Appendix

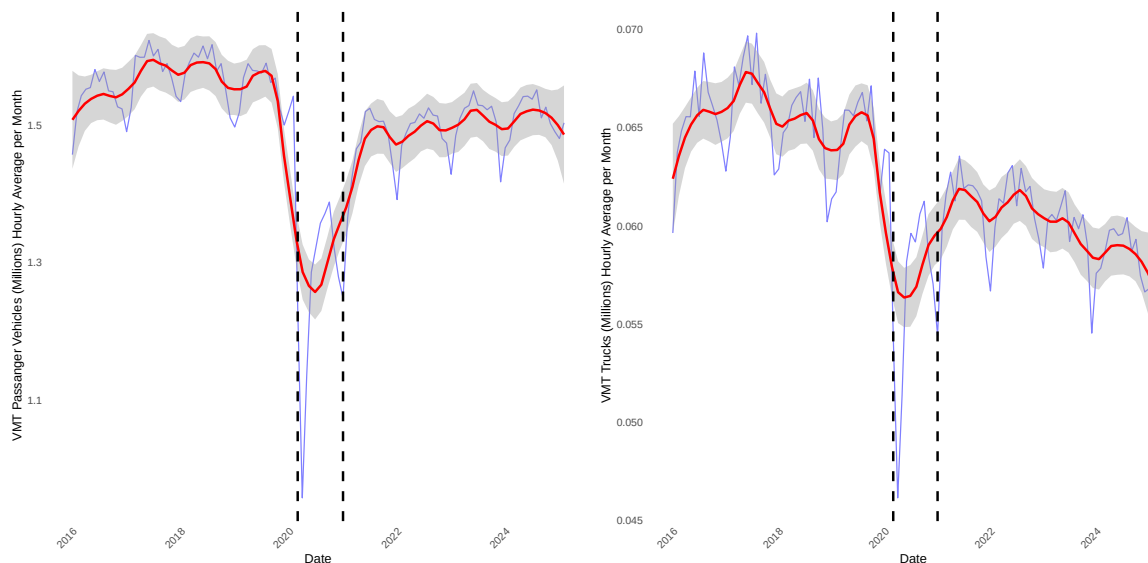
A Robustness Tables and Figures

Table A.1: Major Freeways by County in California

County	Freeways
Los Angeles	I-5, I-10, I-105, I-110, I-210, I-405, I-605, I-710, SR-1, SR-2, SR-14, SR-60, SR-91
San Francisco	I-80, I-280, US 101, SR-1
Orange	I-5, I-405, SR-22, SR-55, SR-57, SR-73, SR-91, SR-133, SR-241
Santa Clara	I-280, I-680, US 101, SR-17, SR-85, SR-87, SR-237, SR-152
San Diego	I-5, I-8, I-15, I-805, SR-52, SR-54, SR-56, SR-75, SR-94, SR-125, SR-163

Note: The freeway data listed here is derived from Caltrans' Performance Measurement System (PeMS). While PeMS includes data for the entire California freeway network, data quality may vary depending on sensor coverage and performance. However, given the urban nature of the counties analyzed, sensor density and data reliability are generally high in these areas.

Figure A.1. VMT for Passenger Vehicles and Trucks from January 2016 to February 2025

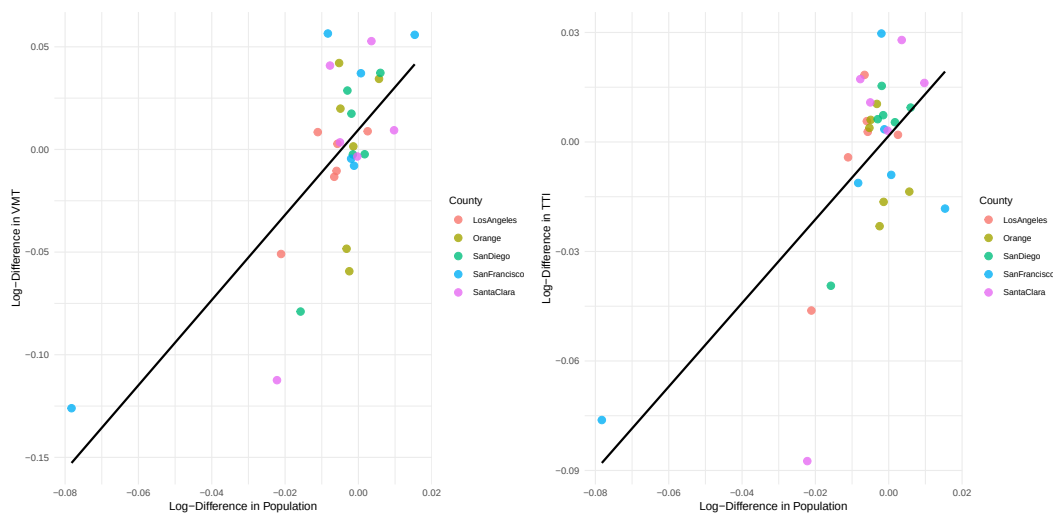


(a) Vehicle Miles Traveled for Passenger Vehicles

(b) Vehicle Miles Traveled for Trucks

Notes: This figure shows the evolution of Vehicle Miles Traveled for passenger vehicles and trucks from January 2016 to February 2025, measured as the monthly average per hour. The blue line represents the raw data. The red line is a LOESS smoother with a 95% confidence interval. Vertical dashed black lines indicate the start and end of the lockdown period.

Figure A.2. Yearly Change in Traffic and Congestion vs. Change in Population



(a) Yearly Change in VMT

(b) Yearly Change in TTI

Notes: Yearly approximated percentage change in VMT and TTI plotted against the yearly approximated percentage change in population for each county. VMT and TTI data are from PeMS; population estimates are from the ACS. Each dot represents a county-year observation.

Table A.2: Differences in Traffic Before and After the Pandemic

	All Data	TTI (3pm6pm)	Weekends
	(1)	(2)	(3)
Pandemic After	−0.029*** (0.001)	−0.065*** (0.002)	−0.007*** (0.001)
Constant	0.018*** (0.0004)	0.197*** (0.001)	−0.036*** (0.001)
Observations	181,655	30,280	51,695
R ²	0.011	0.050	0.001
Adjusted R ²	0.011	0.050	0.001

Note:

*p<0.1; **p<0.05; ***p<0.01

Note: This table shows the difference in means of $\log(TTI)$ before and after the pandemic. Column 1 includes all hourly data from 2018 to 2022, column 2 focuses on rush hour data (3 PM to 6 PM), and column 3 analyzes weekend data.